

IAN TAI AHN

Software Engineer, Data Science M.S

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SUMMARY

Software Engineer with 3+ years of professional experience delivering production systems in high-compliance defense environments (F-35 program), plus an M.S. in Data Science. Independently designs, builds, and deploys production ML systems end-to-end, including a Retrieval-Augmented Generation service iterated from a fully local model deployment to hosted inference APIs (Jina, Groq), with async request handling, cold-start mitigation, and live cloud deployment on Render. Currently serves as Technical Lead and Scrum Master for a mission-critical platform spanning 30+ distributed microservices on the F-35 program. Active Secret clearance. U.S. Citizen.

EXPERIENCE

Department of the Air Force – Hill Air Force Base, UT, Computer Scientist (1550) **Aug 2023 – Present**

Technical Lead & Scrum Master – Virtual Aircraft Modernization & File Sync **Feb 2026 – Present**

- Serve as Scrum Master for a cross-functional team leading Agile ceremonies, sprint planning, and backlog prioritization while maintaining ~70% technical ownership: writing/reviewing code for virtual aircraft modernization and planning/implementing the file synchronization solution for the File Sync initiative.
- Own spike planning, stakeholder alignment, and sprint execution for a file synchronization initiative across several bare-metal servers using PowerShell, Bash, and open-source tooling.
- Coordinate cross-functionally with several engineering teams to align delivery across workstreams; proactively clear blockers and translate ambiguous requirements into sprint-ready work.

Associate Technical Lead – Virtual Aircraft Modernization **Aug 2025 – Feb 2026**

- Delivered full refactor of the React component library for a virtual field damage and repair entry application on the F-35 Lightning II program, eliminating duplicated components and establishing maintainable architecture.
- Contributed across the full stack (React frontend, Unity WebGL via react-unity-webgl, and PostgreSQL backend) while onboarding developers and driving high-complexity cross-functional coordination.

Platform Engineer – Kubernetes Service Mesh **Apr 2024 – Aug 2025**

- Refactored Helm charts and authored Robot Framework tests to improve Kyverno policy enforcement; maintained service mesh health (routing, logging, monitoring, compliance) via Prometheus, Splunk, and Grafana, and supported CI/CD pipelines using Flux and Ansible.

Test Automation Engineer – F-35 Health Systems **Aug 2023 – Apr 2024**

- Built Python data pipelines and automated test frameworks for F-135 turbofan engine health monitoring, transforming tabular data across regression cycles and reducing a 30-minute manual process to seconds (~10 hours/month saved).

EDUCATION

M.S., Data Science (Graduate Certificate, Computational Data Science & ML) | Weber State University | **GPA 3.85** | 2025

B.S., Computer Science | Weber State University | **GPA 3.93** | 2023

PROJECTS

Lightweight RAG Pipeline & Deployment | *Python, FastAPI, React, FAISS, Jina Embeddings/Reranker API, Groq LLM API, Docker*
<https://github.com/IanTaiAhn/rag>

- Designed, built, and deployed an end-to-end RAG service with custom chunking, FAISS vector storage, and MMR-based diverse retrieval, using a FastAPI backend and React frontend live-hosted via Render and GitHub Pages.
- Migrated from locally-hosted embedding, reranking, and generation models to hosted inference APIs, cutting backend dependencies by over half; added async endpoints and a warm-up hook to reduce cold-start latency, and documented architecture trade-offs and rate-limiting behavior from real deployment.

TECHNICAL SKILLS

Languages & Data: Python, Java, SQL, PostgreSQL, JavaScript/TypeScript, Shell/PowerShell/Bash

NLP & AI: custom RAG architectures, LLM APIs (Anthropic Claude, Groq, OpenAI), prompt engineering, structured output generation, Pydantic, pandas, NumPy

Cloud & Platform: Kubernetes, Docker, Helm, Ansible, Kyverno, Prometheus, Grafana, Splunk, CI/CD

Frameworks & Tools: FastAPI, React, REST APIs, Git, Agile/Scrum, JIRA, Confluence, Robot Framework